

TASHFEEN AZIZ

AI Engineer | LangGraph & RAG Systems | Generative AI | Python Backend

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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Agentic AI, and backend development with hands-on experience building production-ready AI systems using Python, FastAPI, LangGraph, Retrieval-Augmented Generation (RAG), and modern Large Language Models. Experienced in developing AI agents, scalable REST APIs, vector databases, and full-stack AI applications through industry internships and real-world projects. Passionate about solving real business problems with AI while continuously learning and adopting emerging technologies.

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | NetsolTech, Lahore

June 2026 – July 2026

- Developed a production-ready multi-agent AI assistant using LangGraph, FastAPI, ChromaDB, and Google Gemini.
- Built Retrieval-Augmented Generation (RAG) pipelines supporting document-based question answering for PDF, DOCX, and TXT files.
- Designed and developed scalable REST APIs to integrate AI agents, document retrieval, and external services into a unified platform.
- Implemented Text-to-SQL analytics, Google Calendar integration, Tavily web search, and real-time streaming APIs.
- Collaborated with senior engineers to build, test, and improve production-ready AI solutions.

AI/ML Trainee | NetsolTech, Lahore

July 2025 – December 2025

- Completed 6-month structured AI/ML program covering computer vision, deep learning, and production ML workflows
- Trained and evaluated ML/DL models using Python, TensorFlow, and PyTorch on real-world datasets
- Applied end-to-end pipeline development: data preprocessing → model training → evaluation → optimization

AI/ML Intern | Stameta.ai, Islamabad

June 2025 – July 2025

- YOLO-based object detection for real-world industry applications; model integration and deployment workflows

PROJECTS

NetsolChatbot — AI Assistant [↗](#)

2026

Built a production-ready multi-agent AI assistant using LangGraph, FastAPI, ChromaDB, React, and Google Gemini. Implemented Retrieval-Augmented Generation (RAG), Text-to-SQL analytics, Google Calendar integration, Tavily web search, semantic document retrieval, and real-time streaming APIs.

AI Voice Khata — WhatsApp Voice Bookkeeping [↗](#)

2026

Developed a WhatsApp-native AI bookkeeping solution that converts Urdu and Roman Urdu voice messages into structured financial records using OpenAI Whisper and GPT-4o. Built scalable FastAPI backend services, integrated WhatsApp Business APIs, and automated PDF invoice generation with Supabase for transaction storage.

CryptoChat — RAG Crypto Assistant [↗](#)

2025

Built a full-stack AI crypto assistant integrating LangChain, LLaMA3, Binance APIs, and Retrieval-Augmented Generation (RAG) to deliver context-aware investment insights using live market data.

HelmetEye — AI Traffic Surveillance (FYP) [↗](#)

2024–2025

Developed an AI-powered traffic surveillance system using YOLOv12, OpenCV, and OCR for real-time helmet detection and automatic vehicle number plate recognition with a FastAPI backend and React frontend.

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, RAG Systems, Tool-Calling, Agentic AI, Prompt Engineering

LLMs: Google Gemini, GPT-4o, LLaMA3, Groq, HuggingFace, OpenAI Whisper

Vector DBs: ChromaDB, Qdrant, Pinecone, Hybrid Search (BM25 + RRF), RAGAS Evaluation

Backend & DB: Python, FastAPI, SQLite, PostgreSQL, Supabase, Git

CV & ML: YOLOv12, OpenCV, TensorFlow, PyTorch, Scikit-learn, MLflow

EDUCATION & CERTIFICATIONS

BS Information Technology | University of Kotli, AJK

2021 – 2025

Certifications: AI/ML 6-Month — NetsolTech · Python for Everybody — Coursera · Intro to SQL — Coursera

Leadership: Lead — GBSO Coding Club (2024–2025) · Tech Lead — Soft-Tech Community (2024–2025)